

GNN–BERT for Understanding Context from Music

S. M. Noiem Uddin

Department of Computer Science and Engineering

Course: Neural Networks (CSE425)

Section: 01

Student ID: 18101323

Email: *s.m.noiem.uddin@g.bracu.ac.bd*

Abstract—Music context is distributed across acoustic structure and language. This paper presents a three-stage study of text-only, audio-graph, and graph-text models for music understanding. A DistilBERT baseline is evaluated on a MusicCaps caption-to-tag proxy task. FMA-small tracks are represented as graphs of five-second audio segments for genre classification with GraphSAGE, and compared with a mel-spectrogram CNN. A graph-text model predicts a fixed multi-label genre/tag vocabulary from FMA metadata text and audio graphs, comparing BERT-only, GNN-only, early-concatenation, and cross-attention fusion. On the reported test runs, the CNN obtains macro-F1 0.4544 versus 0.4323 for GraphSAGE. In the multimodal ablation, graph-only obtains the best macro-F1 (0.0427) and mean PR-AUC (0.1340), while cross-attention obtains the best micro-F1 (0.3546). The results show that graph structure supplies useful signal for the selected FMA labels, but also expose substantial generalization difficulties for the multi-label fusion setting.

Index Terms—music understanding, GraphSAGE, BERT, multimodal learning, audio graphs, multi-label classification

I. INTRODUCTION

Music context includes timbre, rhythm, harmony, genre, mood, and natural language descriptions. Audio-only sequence models capture local patterns but do not explicitly represent relations between repeated or similar segments. Conversely, a language encoder captures contextual semantics but cannot directly observe acoustic structure. We study whether a graph representation of audio segments and a BERT representation of metadata text can provide complementary information.

We consider three complementary settings. A DistilBERT baseline measures text-only performance on a caption-to-tag proxy task. A GraphSAGE model performs single-label genre classification from audio-segment graphs and is compared with a mel-spectrogram CNN. Finally, a graph-text model performs multi-label genre/tag prediction through a four-condition ablation. Results from synthetic data are used only to validate the implementation and are not included as experimental evidence.

II. RELATED WORK

The language branch is based on the contextual representation strategy of BERT [1]; the implementation uses the smaller DistilBERT configuration for the text-only baseline [2]. GraphSAGE learns node representations by sampling and aggregating neighborhood information [3]. The audio features use standard short-time spectral analysis and mel-spectrogram conventions [4]. FMA provides the audio and metadata basis for genre classification and fusion [5], while MusicCaps supplies

natural-language descriptions for the caption-to-tag proxy experiment [6].

III. DATA AND PREDICTION TARGETS

A. Datasets and targets

The text-only experiment uses real MusicCaps captions and a proxy tag target generated from the captions. This target is not a human-annotated MusicCaps tagging benchmark, so its score is reported separately from the FMA experiments. The audio and multimodal experiments use FMA-small. Genre classification is single-label over eight top-level genres: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock.

The multimodal experiment uses the complete FMA metadata source and constructs a fixed multi-label vocabulary. Normalized genres are included first and frequent normalized tags fill the remaining positions up to the configured maximum of 100 labels. The target is multi-hot: a track is positive for its normalized top genre and any selected normalized tags. The exact vocabulary is persisted by the preparation pipeline and reused across all four ablation conditions.

B. Splits and leakage control

All partitions are artist-disjoint: every track from an artist from one artist belong to exactly one of train, validation, or test. The fusion manifest is an inner join on `track_id` between the graph manifest and prepared text metadata, so failed graph construction and rows without usable text are excluded consistently. Feature scaling is fitted on training segments and shared across splits. These choices prevent direct artist leakage and prevent test statistics from influencing graph features.

IV. METHOD

A. Audio segmentation and graph construction

Audio is loaded as mono at 22,050 Hz and partitioned into five-second segments. Each segment node contains MFCC means and standard deviations, first- and second-order MFCC deltas, chroma statistics, spectral contrast, tonnetz, spectral centroid, bandwidth, rolloff, zero-crossing rate, RMS energy, and spectral-flatness statistics. The resulting feature dimension is 182. Temporal-adjacency edges connect consecutive segments. Additional undirected similarity edges are added when

the cosine similarity of the complete audio feature vectors is at least 0.75.

The graph encoder projects node features to a 256-dimensional hidden space, applies three residual GraphSAGE layers with normalization and dropout, and concatenates mean, maximum, and standard-deviation global readouts. A linear head maps the graph representation to the target labels. The corresponding convolutional baseline applies a compact CNN to a 128-bin log-mel spectrogram.

B. Text and fusion models

The text-only baseline uses a DistilBERT encoder with a linear multi-label head. The multimodal model passes only non-target FMA metadata text to BERT: album information, album title/type, artist biography/location/members, composer, track information, and track title. Genre/tag fields, identifiers, dates, URLs, and split metadata are excluded from the text input.

Let g denote the graph readout and H the BERT token sequence. The cross-attention condition uses a graph-derived query:

$$A = \text{softmax} \left(\frac{(gW_Q)(HW_K)^T}{\sqrt{d}} \right), \quad z = [g; AHW_V]. \quad (1)$$

The early-concatenation condition replaces the attended text vector with the BERT pooled output. The text-only and graph-only conditions remove the other branch. All multimodal conditions use binary cross-entropy with logits, and predictions are thresholded at 0.5 for F1.

C. Evaluation

Macro-F1, micro-F1, and mean average precision (reported by the implementation as mean PR-AUC) are reported for multi-label prediction. Accuracy and test loss are additionally reported for genre classification. Checkpoints are selected on the validation split, and all reported scores are computed on held-out test data.

V. EXPERIMENTAL SETUP AND REPRODUCIBILITY

A. Data preparation and validation

The implementation was validated first on controlled synthetic graphs with six labels, random node features, deterministic splits, and serialized graph manifests. This preliminary check verified tensor shapes, checkpoint serialization, evaluation, and qualitative-output generation; synthetic scores are not used in the analysis below.

The reported experiments use a stable 80/10/10 split for the MusicCaps caption proxy and artist-disjoint train/validation/test partitions for FMA. The audio pipeline constructs metadata, segment graphs, and cached mel-spectrograms separately. Feature extraction was expanded from MFCC-only representations to include harmonic, spectral, rhythm, and energy statistics; normalization is fitted on training segments before similarity edges are constructed. The multimodal pipeline scans and normalizes genres and tags, selects a deterministic vocabulary, creates non-target descriptive text, and joins graph and text

records by track identifier. These procedures provide a common data contract for the ablation models. The resulting preparation and validation stages are summarized in Table I.

B. Configuration and reproducibility

Unless overridden by a task command, experiments use seed 42, batch size 8, learning rate 3×10^{-4} , weight decay 10^{-4} , dropout 0.2, three GraphSAGE layers, 256-dimensional graph and text projections, gradient clipping at 1.0, a factor-0.5 learning-rate scheduler, and early stopping with patience 12. The text baseline uses DistilBERT, maximum sequence length 128, batch size 8, learning rate 2×10^{-5} , and the documented real run. The graph and multimodal experiments use the same seed/configuration contract; multimodal runs are capped at 20 epochs and select the best validation Macro-F1 checkpoint. The primary fusion run uses positive weighting, and every ablation uses the same fixed label order. The consolidated settings are listed in Table II.

VI. EXPERIMENTS AND RESULTS

A. Caption-to-tag proxy classification

The DistilBERT test evaluation reports macro-F1 0.9976, micro-F1 0.9991, and PR-AUC 1.0000. Because the experiment uses caption-derived proxy labels, these values should be interpreted as a verification of the text pipeline and not as evidence of general MusicCaps tag understanding. The training trajectory in Fig. 1 reaches a similarly high validation score quickly, consistent with the proxy nature of the target rather than difficult semantic generalization. The held-out values are reported in Table III.

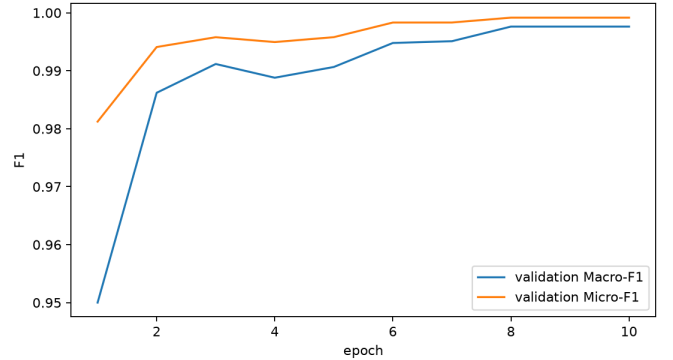


Fig. 1. DistilBERT training and validation F1 curves on the MusicCaps caption-to-tag proxy task.

B. Audio-graph genre classification

Table IV compares GraphSAGE with the mel-spectrogram CNN. The CNN leads GraphSAGE by 0.0222 macro-F1 and 0.0350 PR-AUC, while both models achieve their best per-class F1 on Hip-Hop. GraphSAGE is stronger on International, whereas the CNN is stronger on Experimental, Folk, Hip-Hop, Instrumental, Pop, and Rock.

The per-genre breakdown in Table V is more informative than accuracy alone: GraphSAGE improves Electronic, International,

TABLE I
DATA PREPARATION AND VALIDATION PROCEDURES.

Procedure	Purpose	Evidence
Preliminary validation	Can graph/text training and evaluation run end to end?	src/make_synthetic_data.py, synthetic manifest, checkpoints, and case-study generation
Text baseline	Can contextual text encode the proxy labels?	MusicCaps preparation, DistilBERT checkpoint, metric history, predictions, F1 curve
Data and layout	Which datasets and directory contracts are valid?	FMA and MusicCaps data definitions, processing conventions, and experiment paths
Audio preprocessing	Can real audio become leakage-controlled graphs and cached mels?	FMA metadata, audio features, scaling, graph manifest, split files, failure report, mel cache
Audio modeling	Does graph structure beat a standard spectrogram baseline?	GraphSAGE/CNN checkpoints, learning curves, confusion matrices, predictions, comparison JSON
Multimodal preparation	Can labels and text be built without target leakage?	Scan/select/prepare commands, fixed vocabulary, prepared text, inner-join manifest, validators
Fusion ablation	Which modality and fusion mechanism contributes?	Four checkpoints, histories, predictions, PR curves, t-SNE plots, case studies, comparison analysis

TABLE II
CORE EXPERIMENTAL HYPERPARAMETERS.

Parameter	Value
Seed	42
Audio sample rate / segment	22,050 Hz / 5 s
Mel bins / similarity threshold	128 / 0.75
Graph layers / hidden size	3 / 256
Text model / max length	DistilBERT / 128
Dropout / batch size	0.2 / 8
Learning rate / weight decay	3×10^{-4} / 10^{-4}
Scheduler / early stopping	factor 0.5 / patience 12
Gradient clipping	1.0
Multimodal loss	BCEWithLogitsLoss

TABLE III
DISTILBERT TEST PERFORMANCE ON THE MUSICCAPS CAPTION-TO-TAG PROXY.

Model	Macro-F1	Micro-F1	PR-AUC
DistilBERT	0.9976	0.9991	1.0000

TABLE IV
FMA-SMALL GENRE CLASSIFICATION TEST RESULTS.

Model	Acc.	Macro-F1	Micro-F1	PR-AUC	Loss
Mel-CNN	0.4650	0.4544	0.4650	0.4939	1.4877
GraphSAGE	0.4300	0.4323	0.4300	0.4534	1.7490

and Pop, while the CNN has larger gains on Experimental, Instrumental, and Rock. Both models are strongest on Hip-Hop, suggesting that class separability and class frequency are not uniform across the eight genres. The learning curves in Fig. 2 and Fig. 3 show the optimization behavior behind the final comparison, while the confusion matrices in Fig. 4 and Fig. 5 make class-specific errors visible. The corresponding GraphSAGE learning trajectory is shown in Fig. 3.

TABLE V
PER-GENRE TEST F1 ON FMA-SMALL. BOLD VALUES INDICATE THE STRONGER MODEL FOR EACH GENRE; THE FINAL COLUMN IS CNN MINUS GRAPH SAGE.

Genre	Mel-CNN	GraphSAGE	Δ
Electronic	0.5510	0.5771	-0.0261
Experimental	0.3814	0.2880	+0.0934
Folk	0.2647	0.2513	+0.0134
Hip-Hop	0.7444	0.6900	+0.0544
Instrumental	0.4330	0.3209	+0.1121
International	0.3467	0.4852	-0.1385
Pop	0.3333	0.3491	-0.0157
Rock	0.5809	0.4967	+0.0842
Macro-F1	0.4544	0.4323	+0.0222

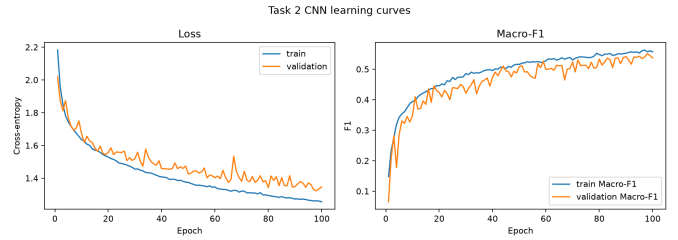


Fig. 2. Mel-spectrogram CNN learning curves.

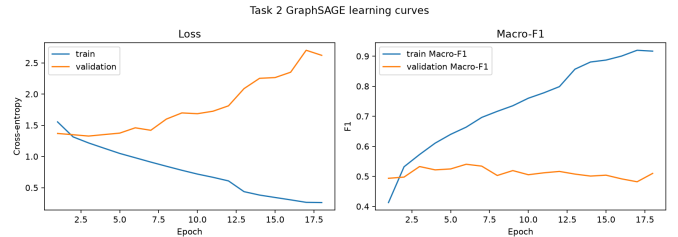


Fig. 3. GraphSAGE learning curves.

C. Multimodal fusion ablation

The four-condition results are shown in Table VI. The graph-only achieves the Mel highest macro-F1 and PR-AUC. Cross-

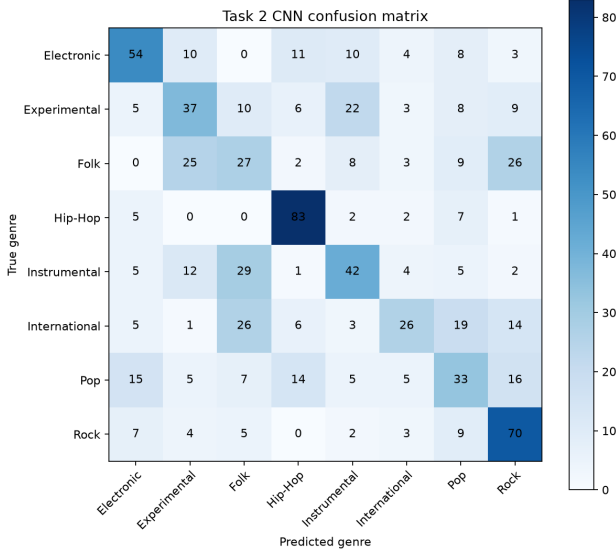


Fig. 4. Mel-spectrogram CNN test confusion matrix.

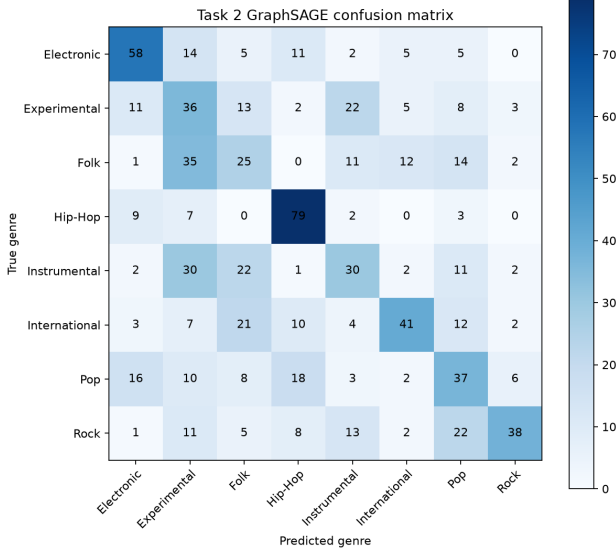


Fig. 5. GraphSAGE test confusion matrix.

attention achieves the highest micro-F1, narrowly exceeding early concatenation. Both fusion conditions substantially improve on BERT-only micro-F1 and PR-AUC, but their macro-F1 remains low, indicating that performance is concentrated in a subset of labels. The gap between training and test metrics in the exported histories also indicates overfitting.

The delta table is useful because the absolute scores alone hide the practical effect of adding audio. Cross-attention gives the largest micro-F1 improvement, while GNN-only gives the largest macro-F1 and PR-AUC improvements. None of the audio-containing conditions minimizes BCE test loss, so the result is not a uniformly dominant fusion model. These

TABLE VI
FMA MULTI-LABEL ABLATION TEST RESULTS.

Condition	Macro-F1	Micro-F1	PR-AUC
BERT-only	0.0111	0.2030	0.0366
GNN-only	0.0427	0.3353	0.1340
Early concat.	0.0403	0.3542	0.1255
Cross-attention	0.0393	0.3546	0.1308

TABLE VII
IMPROVEMENT OVER THE TEXT-ONLY TEST BASELINE. POSITIVE VALUES ARE IMPROVEMENTS FOR F1 AND PR-AUC; FOR LOSS, NEGATIVE VALUES WOULD BE IMPROVEMENTS.

Condition	Δ Macro-F1	Δ Micro-F1	Δ PR-AUC	Δ Loss
BERT-only	0.0000	0.0000	0.0000	0.0000
GNN-only	+0.0315	+0.1323	+0.0974	+1.3756
Early concat.	+0.0292	+0.1512	+0.0889	+0.4578
Cross-attention	+0.0281	+0.1516	+0.0942	+0.6663

baseline-relative changes are reported in Table VII. The visual comparison in Fig. 6 reinforces this metric-dependent ranking.

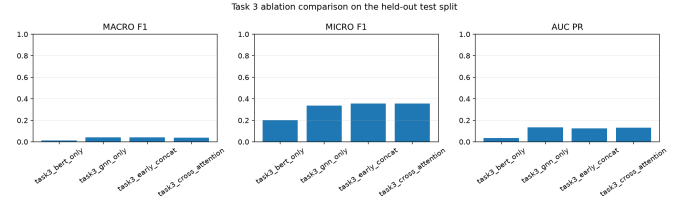


Fig. 6. Multimodal ablation comparison across Macro-F1, Micro-F1, and PR-AUC.

D. Training dynamics and comparative analysis

The training histories provide evidence beyond the final test row. The text baseline reaches its best validation Macro-F1 at epoch 4 in the stored history. The graph-only and fusion conditions reach their best stored validation Macro-F1 at epoch 5, while BERT-only reaches its best at epoch 2. The large train/test gaps, especially for GNN-only and fusion, indicate overfitting and label imbalance rather than a lack of optimization alone.

Relative to BERT-only, the post-ablation analysis reports the following cross-attention changes: Macro-F1 +0.0281, micro-F1 +0.1516, PR-AUC +0.0942, and test loss +0.6663. Thus adding audio improves micro-level decisions and ranking quality, but the improvement is not uniform over labels. The ranking is metric-dependent: GNN-only is best for Macro-F1 and PR-AUC, cross-attention is best for micro-F1, and BERT-only has the lowest test loss. For the selected primary fusion run, Fig. 7 shows the train/validation history and Fig. 8 shows per-label precision-recall behavior. These figures support the overfitting and imbalance interpretation; Table VI remains the primary cross-model evidence. The corresponding checkpoint epochs and test Macro-F1 values are summarized in Table VIII.

TABLE VIII
BEST VALIDATION EPOCHS AND CORRESPONDING TEST MACRO-F1.

Run	Best epoch	Test Macro-F1
DistilBERT proxy	4	0.0409
Text-only	2	0.0111
Graph-only	5	0.0427
Early concat.	5	0.0403
Cross-attention	5	0.0393

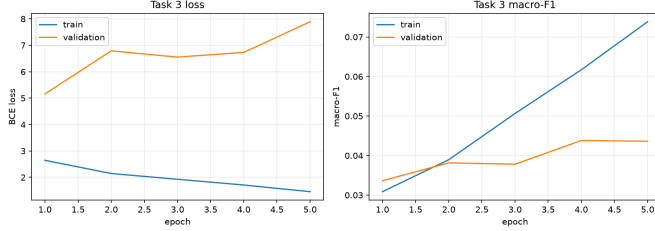


Fig. 7. Cross-attention training and validation history. The widening training-validation gap motivates the overfitting interpretation.

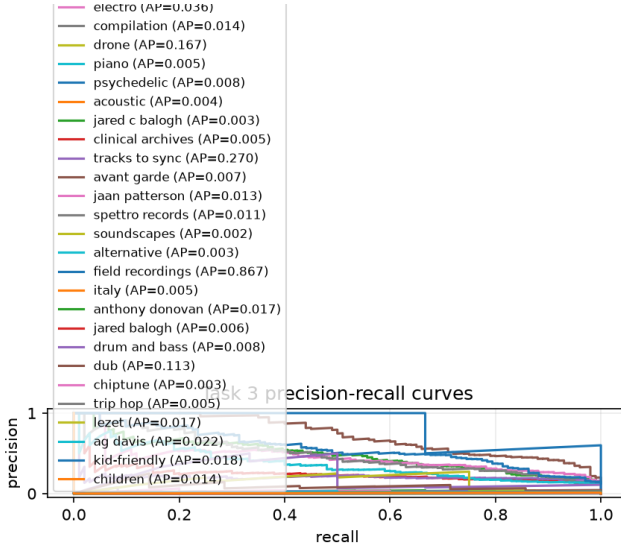


Fig. 8. Cross-attention test precision-recall curves across valid labels.

VII. QUALITATIVE ANALYSIS

The cross-attention t-SNE visualization is shown in Fig. 9. It is used only as a qualitative diagnostic: embedding neighborhoods may suggest label structure, but t-SNE does not replace held-out evaluation. The remaining per-condition plot files remain Additional per-condition diagnostic plots are retained with the experimental artifacts; they are not reproduced here because they do not alter the ablation conclusion.

Three case studies are available for the full cross-attention run. Each case study contains the track identifier, text input, target labels, predicted labels, and prediction scores. These examples are intended to support inspection of graph/text agreement and failure modes. In particular, the low macro-F1 of all multimodal conditions motivates checking rare-label

misses rather than presenting only successful examples.

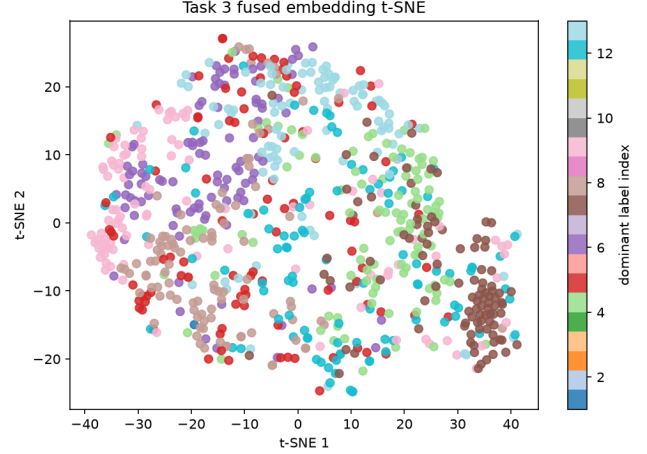


Fig. 9. t-SNE projection of cross-attention fused test embeddings, shown as a qualitative diagnostic rather than a performance metric.

VIII. DISCUSSION

The results provide three practical conclusions. First, the text pipeline is functional, but the caption-proxy construction makes its near-perfect score an upper-bound-like diagnostic rather than a comparable benchmark. Second, the rich segment graph is competitive with a simple audio CNN but does not outperform it on the current FMA-small run; graph topology or the similarity threshold may require further tuning. Third, the multimodal ablation confirms that audio structure is the dominant signal for this label setup. Text fusion improves micro-level decisions over BERT-only, yet does not improve macro-F1 over GNN-only, suggesting label imbalance, limited usable text, or optimization difficulty.

IX. LIMITATIONS AND FUTURE WORK

The present study does not include emotion regression or paired graph-caption retrieval. Audio genre classification, caption-proxy classification, and multimodal tag prediction use different supervision and should not be compared directly. Exact full-data inventory counts are also not available in the committed artifacts.

Several limitations remain. The caption-proxy target is a caption-derived proxy, FMA metadata text is sparse and heterogeneous, and the selected labels are imbalanced. The 0.5 threshold is global rather than label-calibrated. The graph contains rich handcrafted statistics, but the current study does not isolate each feature family or compare multiple similarity thresholds. Finally, the reported runs use one seed; repeated-seed confidence intervals are needed before claiming robust ranking differences.

A. Future-work roadmap

The next experiments follow directly from the preserved findings:

- 1) add per-label validation threshold tuning, calibrated probabilities, and repeated seeds with confidence intervals;
- 2) compare class-balanced, focal, and asymmetric multi-label losses and report rare-label versus frequent-label performance;
- 3) sweep graph similarity thresholds, adjacency variants, pooling strategies, and feature ablations;
- 4) improve text coverage while maintaining strict exclusion of genre/tag targets and artist identity shortcuts;
- 5) add the optional DEAM emotion head as a genuinely multi-task objective;
- 6) implement paired graph-caption contrastive learning and report Caption-to-Audio and Audio-to-Caption Recall@1/5/10;
- 7) expand qualitative analysis with graph paths, attention maps, and failure cases from all four ablations; and
- 8) package exact dataset counts, vocabulary frequencies, split summaries, and preprocessing logs as supplementary reproducibility tables.

[6] A. Agostinelli *et al.*, “Musiclm: Generating music from text,” *arXiv preprint arXiv:2301.11325*, 2023.

X. ARTIFACT MAP FOR REPRODUCTION

The analysis is backed by structured artifacts rather than screenshots alone. The text baseline uses the metrics, test-metrics, predictions, and F1-curve files under `results/task1`. The genre models use both model metric files, predictions, learning curves, confusion matrices, and `task2_model_comparison.json`. The multimodal analysis uses one metrics, prediction, case-study, learning-curve, PR-curve, and t-SNE bundle per condition, plus `task3_ablation_comparison.json` and `task3_ablation_analysis.txt`. Source modules under `src/` define the preprocessing and training contracts, while `README.md` contains the sequential commands needed to regenerate the pipeline.

XI. CONCLUSION

We implemented a reproducible BERT, GraphSAGE, and graph-text fusion pipeline for music context understanding. On the current real-run outputs, the mel-CNN is the strongest genre classifier, while graph representations are the strongest source of macro-level multi-label signal. The ablation results support multimodal fusion as a useful direction, but also make clear that future gains require stronger supervision and improved generalization to rare labels.

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